

ELECTIVE - II

MANAGEMENT INFORMATION SYSTEM (BCA E6006)

TIME-3 hr

FULL MARKS-70

CREDIT-4

The question paper shall consists of two sections: A and B. **Section A** will have eight (08) questions, out of which four (04) questions will be answered and will carry 10 marks each. **Section B** will consists of 10 short answer type questions which will cover the entire syllabus and will carry 30 marks in all., each short-answer type questions carrying 3 marks.

UNIT 1: INTRODUCTION TO MIS

The Technical and Business Perspective, Organization Structure, Evaluation of MIS through Information System, MIS Organization within the Company. [Q-2]

UNIT 2: INFORMATION SYSTEMS FOR DECISION MAKING

Evolution of an Information System, Basic Information Systems, Decision Making and MIS, Decision Assisting Information System, Concepts of Balanced MIS Effectiveness and Efficiency Criteria. [Q-3]

UNIT 3: DEVELOPMENT OF MIS

Methodology and Tools/Techniques for Systematic Identification, Evaluation and Modification of MIS. [Q-1]

UNIT 4: ADVANCED MIS

Concepts, Needs and Problems in Achieving Advanced MIS, DSS. [Q-1]

UNIT 5: PITFALLS IN MIS DEVELOPMENT

Fundamental Weakness, Soft Spots in Planning and Design Problems. [Q-1]

TEXT BOOK:

Murdic, Rose and Clagett- Information Systems for Modern Management, PHI, New Delhi.

REFERENCE BOOK:

Laudon-Laudon- Management Information Systems, Pearson Education, New Delhi.

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Unit 1: Introduction

Attack, Services and Mechanism, Model for Internetwork Security. [Q-1]

Unit 2: Cryptography

Cryptography: Notion of Plain Text, Encryption, Key, Cipher Text, Decryption and cryptanalysis; Public Key Encryption. [Q-2]

Unit 3: Digital signature

Overview, Digital Signatures and Authentication. [Q-2]

Unit 4: Web Security

Requirement, Secure Socket Layer, Transport Layer Security, and Secure Electronic Transactions.

[Q-1]

Unit 5: System Security

Intruders, Viruses and Relate Threats, Firewall Design Principles. [Q-2]

Referential Books:

1. W. Stallings, Networks Security Essentials: Application & Standards, Pearson Education, 2000.
2. W.Stallings, Cryptography and Network Security, Principles and Practice, Pearson Education, 2000.



OPTIMIZATION TECHNIQUES (BCA C6003)

TIME-3 hr

FULL MARKS-70

CREDIT-5

The question paper shall consists of two sections: A and B. **Section A** will have eight (08) questions, out of which four (04) questions will be answered and will carry 10 marks each. **Section B** will consists of 10 short answer type questions which will cover the entire syllabus and will carry 30 marks in all., each short-answer type questions carrying 3 marks.

UNIT 1: ACCOUNTING

Basic of Accounting, Accounting Terminology, Book keeping and Accounting, Accounting Standards, Double Entry System, Classification of Accounts, Rules for Debit and Credit. [Q-2]

UNIT 2: JOURNAL, LEDGER AND TRIAL BALANCE

Meaning of Journal, sub-Journals, Ledger, Trial Balance, Balancing of Ledger accounts. [Q-2]

UNIT 3: CASH BOOK AND SUBSIDIARY BOOKS OF ACCOUNTING

Kinds of cashbook, Purchase daybook, Sales daybook, Bills receivable book, Bills payable book. [Q-1]

UNIT 4: FINANCIAL ACCOUNTS

Trading account,, Profit & Loss account, Adjustments, Balance Sheet, Forms of balance Sheet, Assets and their classification, liabilities and their classification, uses and limitations, fund flow statement, cash flow statement. [Q-1]

UNIT 5: INTRODUCTION TO FINANCIAL MANAGEMENT

Meaning of Finance, Definition of Finance, Types of Finance, Definition of Financial Management,

Scope of Financial Management, Objectives of Financial Management, Approaches to Financial Management (Traditional and Modern Approaches), Functions of Finance Manager and Importance to Financial Management. [Q-2]

TEXT BOOKS:

1. Management Accounting – Manmohan Singh and Goel
2. Financial management- Pandey I. M.

REFERENCE BOOKS:

5. Hanif & Mukherjee-Modern Accountancy, TMH, New Delhi.
6. Maheshwari & Maheshwari- An Introduction to Accountancy, Vikas Publishing House Pvt.Ltd., New Delhi.

PRINCIPLE OF MANAGEMENT (BCA C6002)

TIME-3 hr

FULL MARKS-70

CREDIT-4

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UNIT 1: NATURE OF MANAGEMENT

Meaning, Definition, it's nature purpose, importance & Functions, Management as Art and Science, Concepts of management-Administration-Organization, Management Skills, Levels of Management.

[Q-2]

UNIT 2: EVOLUTION OF MANAGEMENT THOUGHT

Contribution of F.W.Taylor, Henri Fayol, Elton Mayo, Peter Drucker to the management thought. Business Ethics & Social Responsibility: Concept, Shift to Ethics, Tools of Ethics. **[Q-1]**

UNIT 3: FUNCTIONS OF MANAGEMENT: PART-I

Planning – Meaning- Need & Importance, types, Process of Planning, Barriers to Effective Planning. Forecasting- Need & Techniques

Organizing – Elements of organizing & processes:

Types of organizations, Delegation of authority – Need, difficulties

Delegation – Decentralization

Staffing – Meaning & Importance

Direction – Nature – Principles

Communication – Types & Importance **[Q-3]**

UNIT 4: FUNCTIONS OF MANAGEMENT: PART-II

Motivation – Importance – theories

Leadership – Meaning –styles, qualities & function of leader

Controlling - Need, Nature, importance, Process & Techniques

Coordination – Need – Importance **[Q-2]**

TEXT BOOK:

1. Essential of Management – Horold Koontz and Itainz Weibrich- McGrawhills International

REFERENTIAL BOOKS :

2. Management Theory & Practice – J.N.Chandan

3. Essential of Business Administration – K.Aswathapa, Himalaya Publishing House

4. Principles & practice of management – Dr. L.M.Parasad, Sultan Chand & Sons – New Delhi

5. Business Organization & Management – Dr. Y.K.Bhushan

6. Management: Concept and Strategies By J.S. Chandan, Vikas Publishing

7. Principles of Management, By Tripathi, Reddy Tata McGraw Hill

8. Business organization and Management by Talloo by Tata McGraw Hill

OPTIMIZATION TECHNIQUES (BCA C6001)

TIME-3 hr

FULL MARKS-70

CREDIT-4

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UNIT 1: OPERATIONS RESEARCH-AN INTRODUCTION

Definitions of Operations Research, Characteristics of Operations Research Approach. [Q-1]

UNIT 2: LINEAR PROGRAMMING- APPLICATIONS AND MODEL FORMULATION

Graphical Solution Methods of LP Problem., Structure of Linear Programming Model, Advantages of Using Linear Programming, Limitations of Linear Programming, Applications Areas of Linear Programming, General Mathematical Model of Linear Programming Model, Guidelines on Linear Programming Model Formulation, Examples of LP Model Formulation.

[Q-2]

UNIT 3: LINEAR PROGRAMMING- THE SIMPLEX METHOD

Introduction, Standard Form of an LP Problem, Simplex Algorithm (Maximization Case), Simplex Algorithm (Minimization Case). [Q-2]

UNIT 4: TRANSPORTATION & ASSIGNMENT PROBLEM

Introduction, Mathematical Model of Transportation Problem, The Transportation Algorithm, Methods for Finding Initial Solution.

Introduction, Mathematical Model of Statement Assignment Problem, Solution Methods of Assignment Problem. [Q-2]

UNIT 5: PROJECT MANAGEMENT-PERT AND CPM

Introduction, Basic Differences between PERT and CPM, Phases of Project Management, PERT/CPM Network Components and Precedence Relationships, Critical Path Analysis. [Q-1]

TEXT BOOK:

1. J.K Sharma- Operations Research Theory & Applications, 3rd Edn, Macmillan India Ltd., New Delhi-2007.

REFERENCE BOOK:

1. H.A. Taha-Operations Research: An Introduction, Pearson Education, New Delhi, 2006.